

A Short Guide to Proposition C.3

The multimode least-noise moment

Abstract

Proposition C.3 says that the quantum-limited amplifier maximizes every product of decreasing photon-number tests among displacement-covariant amplifiers of the same gain. It allows a trace-nonincreasing map and an arbitrary multimode map that may couple the modes. The proof first reduces to the one-mode stochastic-order theorem of Guță and Matsumoto, then turns the other modes into weighted environments and iterates the one-mode bound. This note explains the normalization, phase twirl, stochastic order, and tensorization argument.

1 Informal meaning

Fix an intensity gain $\gamma > 1$. A displacement-covariant amplifier must send an input displacement z to the amplified displacement $\sqrt{\gamma}z$, but it may add different amounts of noise. The quantum-limited amplifier, whose idler is in the vacuum, adds the least possible noise.

Low photon numbers are rewarded by a decreasing function $w(k)$. Thus the least noisy output has the largest expectation of $w(\hat{N})$. In several modes, Proposition C.3 proves the corresponding statement for product tests

$$W(\hat{\mathbf{N}}) = \prod_{\ell=1}^s w_{\ell}(\hat{N}_{\ell}).$$

The competitor Γ need not factor over the modes; the proof is designed to handle correlations created by a joint channel.

2 Thermal inputs and the reference amplifier

On one bosonic mode, the thermal state with parameter $q \in (0, 1)$ is

$$\tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle\langle k|. \quad (1)$$

Its photon-number distribution is geometric. At gain γ , define

$$q^{(\gamma)} = 1 - \frac{1 - q}{\gamma}. \quad (2)$$

The quantum-limited amplifier maps

$$\tau_q \longmapsto \tau_{q^{(\gamma)}}.$$

Since

$$q^{(\gamma)} - q = \frac{(\gamma - 1)(1 - q)}{\gamma} > 0,$$

the output geometric law is broader than the input law, as amplification requires.

For s modes, choose $q_1, \dots, q_s \in (0, 1)$ and write

$$\tau = \bigotimes_{\ell=1}^s \tau_{q_\ell}, \quad \tau^{(\gamma)} = \bigotimes_{\ell=1}^s \tau_{q_\ell}^{(\gamma)}. \quad (3)$$

3 Decreasing tests and stochastic order

A probability law P on $\mathbb{Z}_{\geq 0}$ stochastically dominates Q if

$$\sum_k f(k) P_k \geq \sum_k f(k) Q_k$$

for every bounded increasing function f . Equivalently,

$$\sum_k w(k) P_k \leq \sum_k w(k) Q_k \quad (4)$$

for every bounded nonnegative decreasing function w .

The direction is easy to remember: the stochastically larger law places more mass at high photon numbers, so it scores less under a test that rewards low photon numbers. Proposition C.3 uses this scalar order for the photon-number distributions of phase-invariant states; it does not assert an operator inequality between density matrices.

4 Statement of Proposition C.3

Proposition C.3 (rephrased). *Let Γ be a completely positive trace-nonincreasing map on the s -mode trace class satisfying*

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^* \quad (z \in \mathbb{C}^s).$$

If each $w_\ell : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}_+$ is bounded and decreasing, then

$$\mathrm{Tr} \left[\Gamma(\tau) \prod_{\ell=1}^s w_\ell(\hat{N}_\ell) \right] \leq \mathrm{Tr}[\Gamma(\tau)] \mathrm{Tr} \left[\tau^{(\gamma)} \prod_{\ell=1}^s w_\ell(\hat{N}_\ell) \right]. \quad (5)$$

The factor $\mathrm{Tr} \Gamma(\tau)$ is essential. It makes the inequality homogeneous and allows Γ to be a probabilistic operation or a weak limit of channels that has lost trace at infinity.

5 Proof: the one-mode case

Step 1: phase twirling changes neither side

Let $R_\theta = e^{i\theta\hat{N}}$. Replace Γ by

$$\Gamma^{\mathrm{ph}}(X) = \frac{1}{2\pi} \int_0^{2\pi} R_{-\theta} \Gamma(R_\theta X R_\theta^*) R_{-\theta}^* d\theta.$$

The thermal input and $w(\hat{N})$ commute with every R_θ , so both traces in (5) are unchanged. The twirl preserves complete positivity, trace nonincrease, and displacement covariance, and it makes the map phase covariant. Independent phase twirls are used in the multimode case.

Phase covariance is what lets the proof speak about the output photon-number law and makes the idler state diagonal in its number basis.

Step 2: covariance makes the success probability constant

The positive output-trace functional has the form

$$\mathrm{Tr} \Gamma(X) = \mathrm{Tr}(AX), \quad 0 \leq A \leq I.$$

Taking traces in the displacement covariance relation shows

$$D(z)^* A D(z) = A \quad (z \in \mathbb{C}).$$

The Weyl representation is irreducible, so Schur's lemma gives

$$A = cI, \quad \mathrm{Tr} \Gamma(X) = c \mathrm{Tr} X, \quad 0 \leq c \leq 1. \quad (6)$$

If $c = 0$, positivity forces $\Gamma = 0$ and the desired inequality is trivial. If $c > 0$, the normalized map

$$\Lambda = c^{-1} \Gamma$$

is a trace-preserving, phase-insensitive, displacement-covariant amplifier of gain $\sqrt{\gamma}$.

Step 3: apply the least-noise stochastic theorem

The Weyl-operator characterization of such amplifiers writes

$$\Lambda^*(D(\xi)) = \mathrm{Tr}[\sigma D(-\sqrt{\gamma-1} \bar{\xi})] D(\sqrt{\gamma} \xi) \quad (7)$$

for an idler state σ . Phase covariance makes σ diagonal in the number basis. Guță and Matsumoto's least-noise result says that, on the thermal input τ_q , the photon-number law of $\Lambda(\tau_q)$ stochastically dominates the law of the vacuum-idler output $\tau_{q(\gamma)}$.

Applying the decreasing-test formulation (4) gives

$$\mathrm{Tr}[\Lambda(\tau_q) w(\hat{N})] \leq \mathrm{Tr}[\tau_{q(\gamma)} w(\hat{N})].$$

Multiplying by $c = \mathrm{Tr} \Gamma(\tau_q)$ proves (5) for one mode.

The stochastic minimality in this step is the specialized external input. The phase twirl, constant-trace reduction, normalization, and conversion to the conic inequality are internal parts of the proof.

6 Proof: the multimode reduction

One cannot simply multiply the one-mode inequalities because Γ may couple all modes. Instead, the proof packages the other modes into a weighted environment.

If one of the weights is identically zero, both sides of (5) vanish. Otherwise rescale each weight by its positive supremum; by homogeneity it is enough to assume

$$0 \leq w_\ell \leq 1.$$

Set

$$W_{>1} = \prod_{\ell=2}^s w_\ell(\hat{N}_\ell), \quad \tau_{>1} = \bigotimes_{\ell=2}^s \tau_{q_\ell},$$

and define a one-mode map

$$\Gamma_1(X) = \text{Tr}_{2,\dots,s} \left[(I \otimes W_{>1}^{1/2}) \Gamma(X \otimes \tau_{>1}) (I \otimes W_{>1}^{1/2}) \right]. \quad (8)$$

Every operation in (8) is completely positive. Since $0 \leq W_{>1} \leq I$ and Γ is trace nonincreasing, Γ_1 is trace nonincreasing. It inherits displacement covariance in the first mode; after the independent phase twirl, it also inherits phase covariance. The one-mode bound therefore yields

$$\text{Tr}[\Gamma(\tau)W] \leq \text{Tr}[\tau_{q_1^{(\gamma)}} w_1(\hat{N}_1)] \text{Tr}[\Gamma(\tau)W_{>1}]. \quad (9)$$

Equation (9) removes the first weight and replaces it by the corresponding quantum-limited expectation. Repeat the same construction for modes $2, \dots, s$. After the last mode, the remaining term is $\text{Tr} \Gamma(\tau)$, while the extracted scalar factors multiply to

$$\prod_{\ell=1}^s \text{Tr}[\tau_{q_\ell^{(\gamma)}} w_\ell(\hat{N}_\ell)] = \text{Tr}[\tau^{(\gamma)}W].$$

This is precisely (5).

7 The particular witness used later

For the orbital Gaussian model, let

$$\tau = \sum_{\mathbf{k}} t_{\mathbf{k}} |\mathbf{k}\rangle\langle\mathbf{k}|, \quad \tau^{(\gamma)} = \sum_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)} |\mathbf{k}\rangle\langle\mathbf{k}|.$$

Appendix C introduces

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle\langle\mathbf{k}|. \quad (10)$$

Because both thermal laws are products,

$$\sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} = \prod_{\ell} \sqrt{\frac{1 - q_\ell}{1 - q_\ell^{(\gamma)}}} \left(\frac{q_\ell}{q_\ell^{(\gamma)}} \right)^{k_\ell/2}.$$

Each factor is positive, bounded, and decreasing. It also tends to zero as $k_\ell \rightarrow \infty$, so W_γ is compact and injective. Proposition C.3 applies to this witness, and direct calculation gives

$$\text{Tr}(\tau^{(\gamma)}W_\gamma) = \text{Tr}(\tau W_\gamma^{-1}) = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} = B_{\text{orb}}(\gamma, p). \quad (11)$$

Thus one witness simultaneously matches the weighted fidelity inequality, the compact-limit topology, and the least-noise moment bound.

8 Dependencies and role in the paper

The logical core is

$$\begin{aligned} &\text{Weyl irreducibility and one-mode least-noise stochastic order} \\ &+ \text{mode-by-mode reduction} \implies \text{Proposition C.3.} \end{aligned}$$

Proposition C.4 applies it to the exactly covariant limit of Følner-averaged orbital channels. Proposition C.6 applies it to a classically weighted quantum limit after first proving that the limit is trace nonincreasing. In both cases, (11) converts the moment estimate into the factor $B_{\text{orb}}(\gamma, p)$.

Primary source for the one-mode stochastic theorem. M. Guță and K. Matsumoto, “Optimal cloning of mixed Gaussian states,” *Physical Review A* **74**, 032305 (2006), [arXiv:quant-ph/0605161](https://arxiv.org/abs/quant-ph/0605161).